

Web and Multimedia Technologies

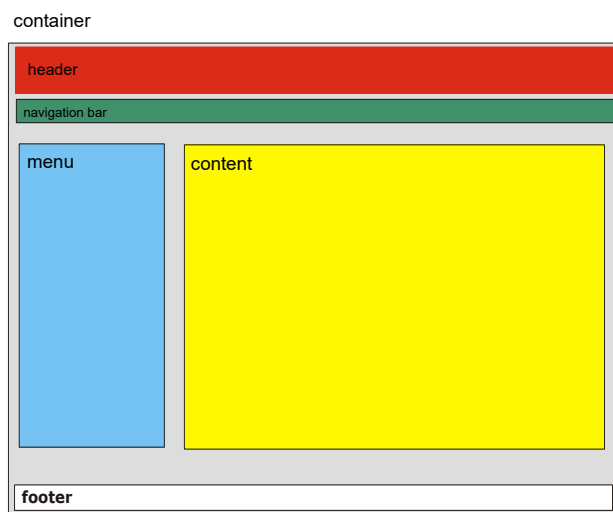
Lab class 4 – Page layout with CSS

Instructions to use the laboratory computers (room B2/B4):

1. If the computer is off, start it with Windows (all computers have a Linux-Windows dual boot)
2. Your username is *cl* followed by the six digits of your student ID; your password is *inf* followed by the last two digits of your student ID (e.g., if the student ID is 123456, the username is *cl123456* and the password is *inf56*)
3. At the end of the class, please log out

Steps:

1. We want to create a layout like the following:



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>A CSS layout</title>
  </head>
  <body>
    <section id="container">
      <header id="header"></header>
      <article id="content"></article>
      <nav id="navbar"></nav>
      <nav id="menu"></nav>
      <footer id="footer"></footer>
    </section>
  </body>
</html>
```

2. Choices: (1) Fixed or liquid layout? (2) Centered or left/right-aligned layout? (3) Which kind of positioning for each box?

Example:

- a. *container* → fixed width, centered, relative positioning
- b. *header* → normal-flow positioning
- c. *content* → normal-flow positioning
- d. *navbar* → absolute positioning
- e. *menu* → fixed width, absolute positioning
- f. *footer* → normal-flow positioning

3. Styles for *body* and *container* (in an external style sheet – e.g., ‘mystyle.css’):

```
body {
  margin: 0px; padding: 0px;
  font-size: 85%;
  font-family: Verdana, sans-serif;
  text-align: center;
  color: black;
  background-color: white;
}

#container {
  padding: 0px;
  margin: 1em auto;
  position: relative;
  width: 630px;
  height: 150px; /* to see it initially... */
}
```

```
text-align: left;
background-color: lightgrey;
border: 1px solid black;
}
```

4. Style for *header*:

```
#header {
  height: 80px;
  background-color: red;
}
```

5. Style for *content*, supposing that the navigation bar is 25 pixels high (to see it, it is necessary to add some content in the *article* with `id="content"`):

```
#content {
  margin-top: 25px;
  margin-left: 129px;
  background-color: yellow;
}
```

6. Style for *navbar*:

```
#navbar {
  position: absolute;
  top: 80px; left: 0px;
  height: 25px;
  width: 100%;
  background-color: lightgreen;
}
```

7. Style for *menu* (to see it, it is necessary to add some content in the *nav* with `id="menu"`):

```
#menu {
  position: absolute;
  top: 111px; left: 0px;
  width: 109px;
  background-color: cyan;
}
```

8. Style for *footer*:

```
#footer {
  height: 20px;
  background-color: white;
}
```

9. Specify padding areas for *content* (e.g., *1em*) and *menu* (e.g., *10px*), the same on all four sides

10. Remove the height definition for *container*

11. Add background images to *body* and *header* (download the files

<https://vision.unipv.it/wmt/lab/csslayout/back.jpg> and

<https://vision.unipv.it/wmt/lab/csslayout/header.jpg>): use the *background-image* property (value `url ("...")`)

12. Define link properties. For example:

```
a {
  text-decoration: none;
  font-weight: bold;
  color: #932f02;
  font-size: 80%;
}
```

```
a:hover {
  text-decoration: underline;
}
```

13. Change to white the *container*'s background color

14. Place some content in the navigation bar (e.g., generic links *Section 1*, *Section 2*, etc.) and “beautify” it:

```
#navbar {
  ...
  border-top: solid white 1px;
  padding-top: 5px;
  background-color: #ffdf71;
}

#navbar a {
  padding-left: 0.7em;
  padding-right: 1em;
  border-right: white solid 1px;
}
```

15. Change to *white* the *content*'s background color and place some content in it (create two paragraphs, for example, copying the text available at <https://vision.unipv.it/wmt/lab/csslayout/content.txt>; also, download the image <https://vision.unipv.it/wmt/lab/csslayout/im.jpg> and, after creating a generic class *right* that defines a right float positioning, place it just at the beginning of the second paragraph, applying the *right* class. Use an *Arial* font (*sans-serif* as a second choice) for paragraphs

16. Include an *h1* heading (e.g., “Long life to CSS”) and an *h2* sub-heading (e.g., “November 10, 2025”) just before the paragraphs, and set their properties; for example:

```
h1 {
  font-family: serif;
  color: red;
  font-size: 1.8em;
  font-weight: normal;
}

h2 {
  margin-top: -1.1em;
  font-family: serif;
  color: #ff6666;
  font-size: 1.3em;
  font-weight: normal;
  font-style: italic;
}
```

17. Change to white the *menu*'s background color and place some content in it; for example:

```
<h3>Archive</h3>
<ul>
  <li><a href="#">Jan 2025</a></li>
  <li><a href="#">Mar 2025</a></li>
  <li><a href="#">May 2025</a></li>
  <li><a href="#">Jul 2025</a></li>
  <li><a href="#">Sep 2025</a></li>
  <li><a href="#">Nov 2025</a></li>
</ul>
```

Also, define the properties for *h3* headings and unnumbered lists:

```
h3 {
  font-family: serif;
  color: red;
  font-size: 1.5em;
  font-weight: normal;
}

ul {
  padding: 0px;
  margin-left: 1.5em;
}
```

18. Place some content in the *footer* (e.g., “© My Website 2025”) and set its properties:

```
#footer {
  height: 20px;
  background-color: #ffdf71;
  border-bottom: solid white 1px;
  text-align: right;
  font-size: 75%;
  padding-top: 0.4em;
  padding-right: 1em;
}
```

19. Define a generic class *it* for italic text, and apply it (for example, by means of a `<span>` tag) to the *footer*’s text “My Website”:

```
.it { font-style: italic }
```

20. Add a left border to *content*:

```
border-left: solid 1px #ffdf71;
```

21. Add to the *header* an *h1* heading with *id*=“*headertitle*” containing the text “The power of CSS”; in the stylesheet, specify that *headertitle* must not be displayed (value *none* for the *display* property... this element is invisible, but it is part of the content!)

22. How can the fixed layout be transformed into a liquid layout? (1) Specify a percentage width (e.g., 80%) for the *container*; (2) in the *header*, use the background image <https://vision.unipv.it/wmt/lab/csslayout/header2.jpg> instead of *header.jpg*, and specify that it must not be repeated (*no-repeat* value for the *background-repeat* property); also, set the background color to *rgb(251, 195, 36)*

23. How to move the menu to the right? Only three simple changes to *content* and *menu* are necessary...